Sales Performance Analysis Report

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Executive Summary

The main goal of this report is to analyze Superstore's performance.
It focuses on finding what increases profit and what causes losses. The purpose is to use data to improve sales strategies and increase overall profit.

The analysis was done using Business Intelligence (BI) tools. Data from CSV files was cleaned and visualized using Power BI. Methods included Trend Analysis for sales growth, Correlation Analysis to determine the impact of discounts on profit, and Categorical Segmentation to compare performance across product lines and regions.

Major Findings:

Regional Leaders: The West Region emerged as the most profitable area, while the

Central Region faced significant margin erosion.

Profit Engines: The Technology category consistently delivers the highest profit

margins, particularly in Phones and Accessories.

Customer Dynamics: The Consumer segment contributes the highest volume, but the

Corporate segment offers more stable per-transaction profitability

Conclusions:

Superstore has strong sales, but the problem is not sales, it is profit.
The company gives too many discounts and has high costs in some product categories.

Instead of focusing only on selling more, the company should focus on selling with better profit margins.

This will increase profit without reducing sales.

Introduction

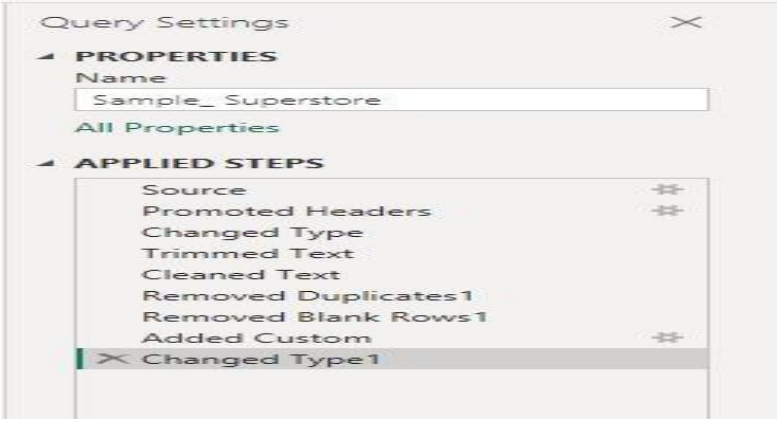
The modern retail environment requires a deep understanding of data to maintain a competitive edge. This report analyzes the operational datasets of the Superstore to evaluate its commercial health. The primary objective is to investigate how variables such as geographical location, product category, and discount strategies influence the overall profitability.

Data Cleaning Process

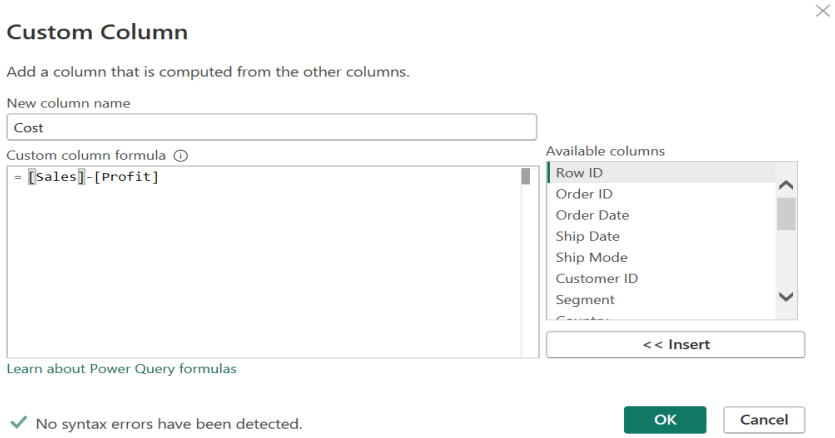
Data Cleaning: Applied 'Trim' and clean functions to the Product Name column to remove extra spaces and non-printable characters, ensuring data consistency.

Data Deduplication: Removed duplicate records from the dataset to ensure each transaction is unique and prevent inflated calculation results

Data Quality: Eliminated blank rows to clean the dataset from empty entries, ensuring a professional and error-free data model.



Created a custom 'Cost' column using a Power Query formula to enable detailed profitability analysis and allowing for more comprehensive financial KPIs in the report and then change the type of cost column to be decimal



Analysis & Discussion

Key Performance indicators

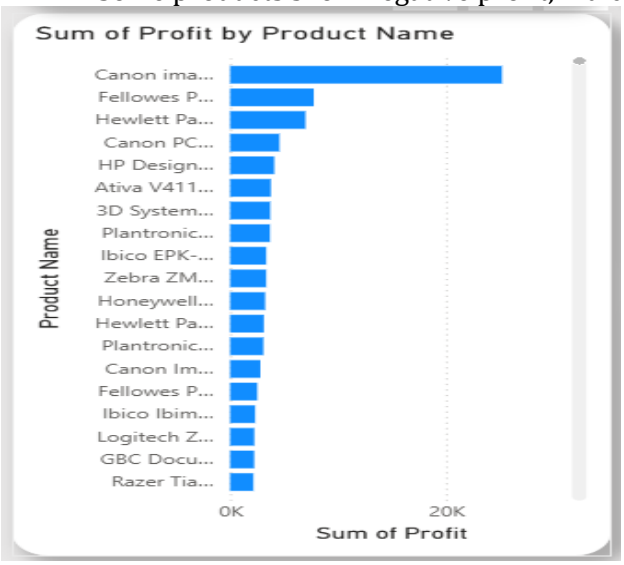
Metric	Value	% of Total
Total Sales	$2.30M	100%
Total Cost	$2.01M	87.4%
Total Profit	$286.4K	12.4%
Total Quantity	38K units	100%

Key Visualization

1.profit by product name

A horizontal bar chart displaying profit contribution from individual products. This visualization reveals which products are generating the highest profit margins and identifies underperforming items.

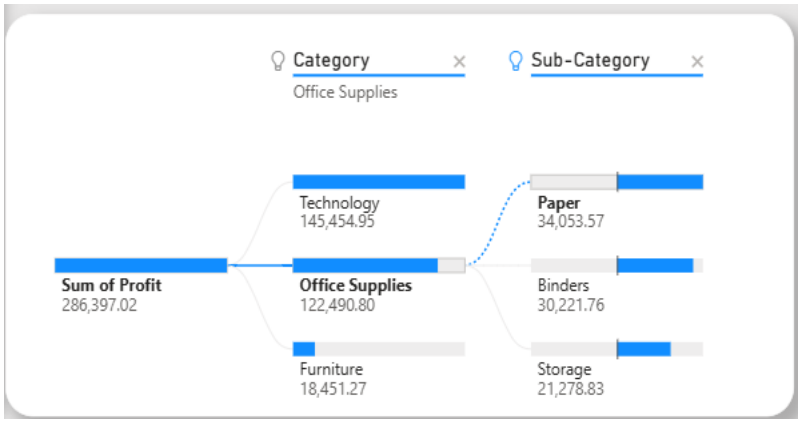
- High-performing products drive significant profit portion
- Some products show negative profit, indicating pricing or cost issues



2.profit by category

A donut chart displaying the distribution of profits across three main product categories.

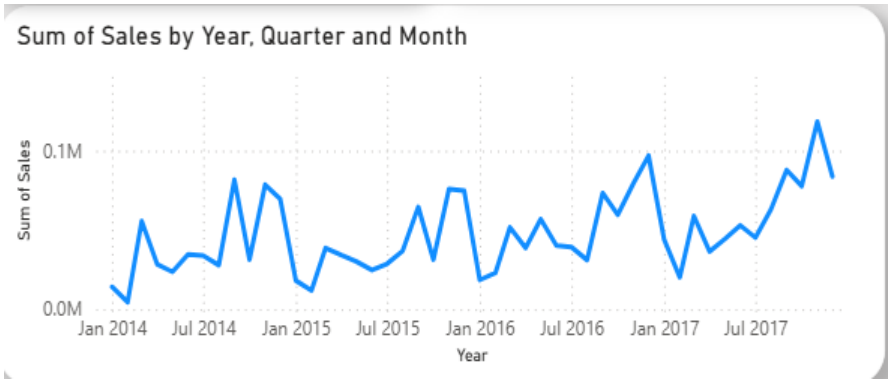
- Technology: 145.1K (50.6% of total profit)
- Office Supplies: 122.5K (42.7% of total profit)
- Furniture: 18.5K (6.4% of total profit)



3.sales by year and month

A line graph depicting sales trends from 2014 to 2017. Key insights include upward growth trend, consistent seasonality with Q4 peaks exceeding 100K, and minimum sales around 50K.

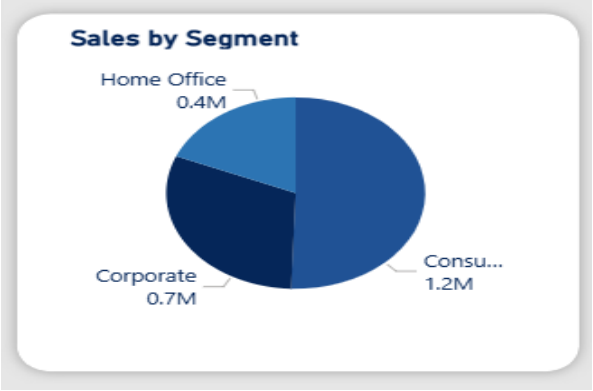
- Clear upward trend in sales volume over 4-year period
- Consistent seasonality with peaks in November-December



4.sales by segment

A pie chart illustrating the distribution of sales across three customer segments.

- Consumer: 1.2M (largest segment, strong B2C presence)
- Corporate: 0.7M (second largest, solid B2B market presence)
- Home Office: Small but emerging segment with growth potential



5.sales by state

A choropleth map of the United States displaying sales distribution across states. This enables regional performance analysis and market penetration assessment.

- Sales distributed across most US states with varying intensities
- Certain regions show higher sales concentration
- Opportunity exists for market expansion in underperforming regions

Data Analysis & Strategic Insights

Profitability Analysis

- Technology dominates profit with 50.6%, making it the core business driver
- Office Supplies provide steady profitability at 42.7%
- Furniture underperforms at 6.4%, indicating need for strategic review
- Product-level variance shows significant opportunity for optimization

Market Segmentation

- Consumer segment represents largest contributor with 1.2 M volume
- Corporate segment shows substantial volume at 0.7M
- Home Office remains underdeveloped but offers growth potential

Conclusion

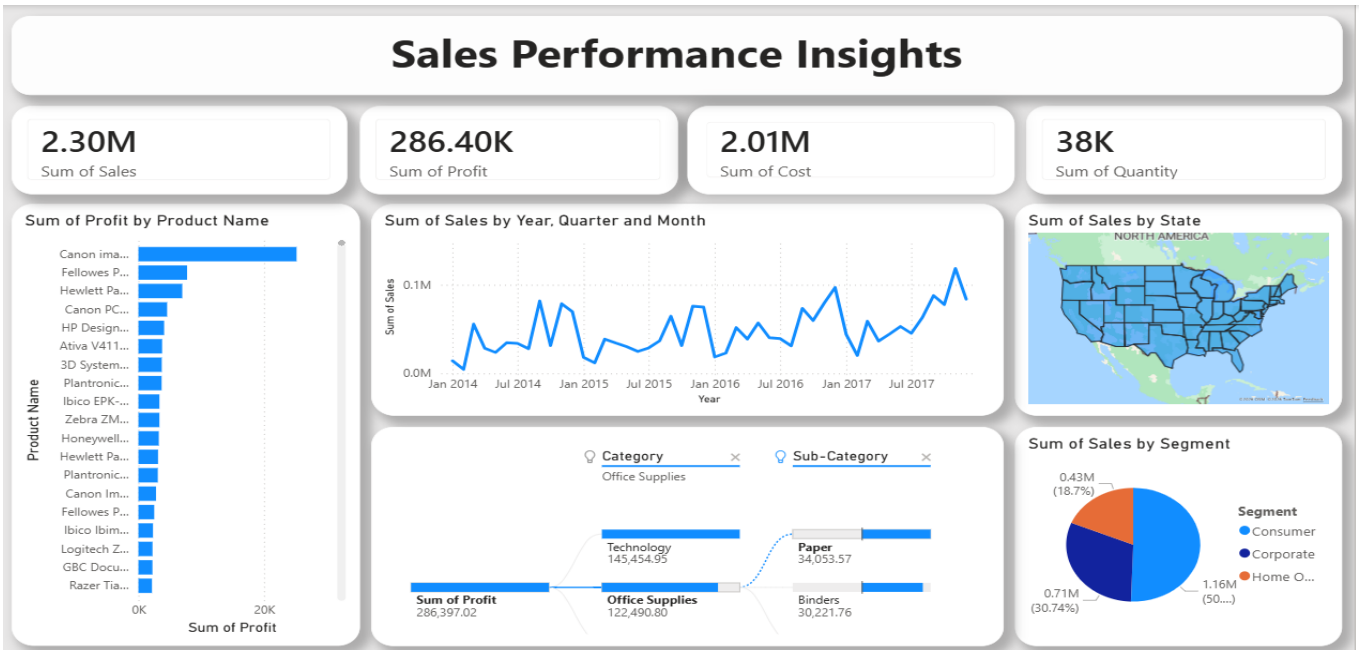
This Power BI dashboard serves as a critical business intelligence tool, providing comprehensive visibility into sales performance across multiple dimensions. The data reveals a healthy business with strong profit margins (12.4%) and consistent growth trends. The technology category drives profitability while other categories offer optimization opportunities.

Recommendations

- Check products that are losing money and decide if we should change price, reduce costs, or stop selling them
- Invest more in the Technology category because it makes high profit
- Improve the Furniture category to make it more profitable
- Focus on weak regions to increase sales there
- Create a plan for Home Office products to match remote work demand
- Use last quarter of the year for better stock and promotions planning

References

The dashboard is built on the Sample Superstore dataset, containing comprehensive sales transaction data including product information, customer segments, geographical locations, and financial metrics spanning 2014-2017 with 38K+ transaction records.


Sales Performance Insights
2.30M
Sum of Sales
286.40K
Sum of Profit
2.01M
Sum of Cost
38K
Sum of Quantity
Sum of Profit by Product Name
Product Name
Canon ima...
Fellowes P...
Hewlett Pa...
Canon PC...
HP Design...
Ativa V411...
3D System...
Plantronic...
Ibico EPK-...
Zebra ZM...
Honeywell...
Hewlett Pa...
Plantronic...
Canon Im...
Fellowes P...
Ibico Ibim...
Logitech Z...
GBC Docu...
Razer Tia...
0K
20K
Sum of Profit
Sum of Sales by Year, Quarter and Month
Sum of Sales
0.1M
0.0M
Jan 2014
Jul 2014
Jan 2015
Jul 2015
Jan 2016
Jul 2016
Jan 2017
Jul 2017
Year
Sum of Sales by State
NORTH AMERICA
Category
Office Supplies
Sub-Category
Technology
145,454.95
Paper
34,053.57
Sum of Profit
286,397.02
Office Supplies
122,490.80
Binders
30,221.76
Sum of Sales by Segment
0.43M
(18.7%)
0.71M
(30.74%)
1.16M
(50....)
Segment
Consumer
Corporate
Home O...